

Assignment 1 — Univariate and Multivariate Statistics

FinTech 545 — Quantitative Risk Management

1 Reading the Shape of a Sample (1.5 points)

`problem1.csv` holds a single series, x .

Predict

Compute the first four moments. From the skewness and the excess kurtosis alone, and before you fit anything, answer:

- a. Which of the distributions from Week 1 could plausibly have produced this sample, and which are ruled out? Name the moment that rules out each one you reject.

The first four moments of the sample are:

- Mean = 0.0011043942883968638
- Variance = $9.624822112193506 \times 10^{-5}$
- Skewness = -0.6698366405653348
- Excess kurtosis = 2.357614233972389

The sample skewness is -0.6698366405653348 , which is negative. This indicates that the distribution is left-skewed, with a longer left tail than the right tail. Based on this negative skewness, I would rule out the Lognormal distribution because the Lognormal distribution is positively skewed. Therefore, the skewness rules out the Lognormal distribution. I would also rule out the Normal distribution because the Normal distribution has zero skewness, while the sample has negative skewness.

The sample has an excess kurtosis of 2.357614233972389, which is positive. This indicates that the sample has heavier tails than a Normal distribution. This provides another reason to rule out the Normal distribution, since the Normal distribution has excess kurtosis equal to zero.

The Student's t distribution has positive excess kurtosis, so its kurtosis is consistent with the sample. However, the Student's t distribution is symmetric and therefore has zero skewness. Since the sample has negative skewness, the Student's t distribution is ruled out by skewness.

The Normal Inverse Gaussian distribution is plausible because it can accommodate nonzero skewness, including negative skewness, as well as positive excess kurtosis. Therefore, based on skewness and excess kurtosis alone, I would consider the Normal Inverse Gaussian distribution a plausible candidate, while the Normal, Lognormal, and Student's t distributions are ruled out by the moments described above.

Fit and reconcile

Fit a Normal to the sample by matching the mean and variance. Then answer:

- b. How many observations fall below the Normal's 1% quantile? How many should there be? Report both numbers.

There are 26 observations below the fitted Normal distribution's 1% quantile. Under the fitted Normal model, we would expect 10 observations below the 1% quantile, since 1% of the 1,000 observations is 10.

- c. If we used this fitted Normal as our model of the series, what would it get wrong? Say where in the distribution the error shows up, and in which direction.

If we used this fitted Normal as our model of the series, we would underestimate the probability of extreme negative outcomes in the left tail of the distribution. Under the fitted Normal model, we would expect only 10 observations below the 1% quantile, but there are actually 26 observations. This shows that the left tail is longer/more extreme than the fitted Normal, so the Normal model would underestimate the frequency of extreme negative values and therefore underestimate left-tail risk.

2 A Regression Whose Errors Are Not Normal (2 points)

`problem2.csv` holds x and y .

Predict

Plot y against x before fitting anything.

- a. Describe what you see in the scatter. What distribution do you expect the error term to follow, and why?

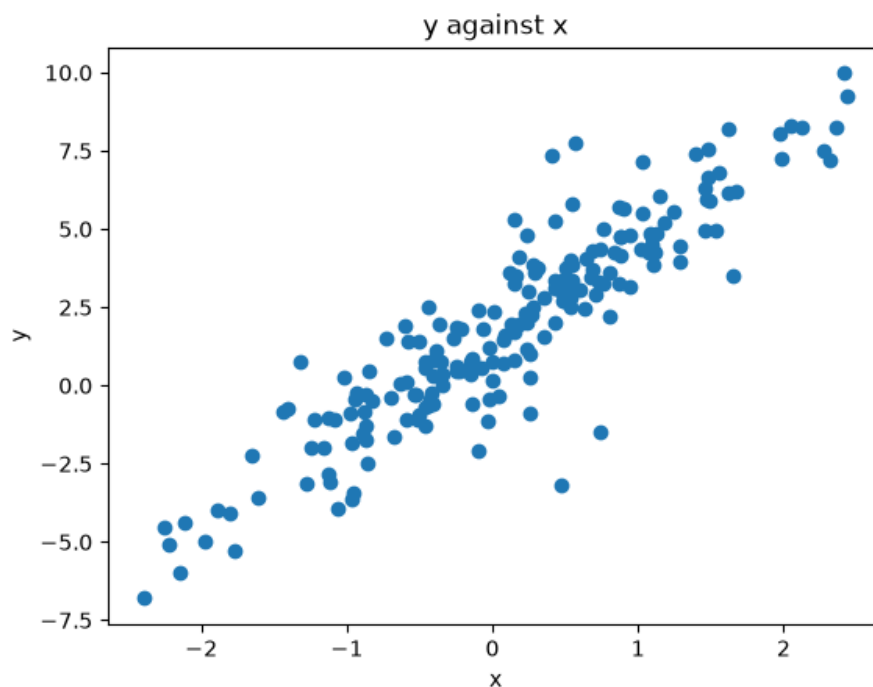


Figure 1: y against x

I notice that as x increases, y also tends to increase, so there appears to be a positive linear relationship between x and y . The points are more concentrated when x is between approximately -1 and 1 , while there are fewer observations outside this range. The spread of the points around the linear trend does not appear to be constant. Therefore, I would expect the error term to have a non-Normal distribution.

- b. Ordinary least squares assumes normally distributed errors. Name which of the seven OLS assumptions this data violates, and predict what that violation will do — to the estimated slope, and separately to the standard error of that slope.

The data violates the OLS assumption that the error terms are normally distributed. This violation does not necessarily bias the estimated slope, as long as the conditional mean-zero assumption still holds, i.e., $E[\varepsilon | X] = 0$. However, non-normal errors may make the usual standard error of the estimated slope unreliable. Consequently, hypothesis tests and confidence intervals based on the conventional standard error may also be unreliable because both rely on the standard error of the estimated slope.

Fit

Estimate three models:

1. OLS, with the standard errors.
2. Maximum likelihood under a Normal error.
3. Maximum likelihood under a Student's t error.

Report α , β , and the fitted error parameters for each. Choose between the models using AICc.

Three models were fitted to the data: ordinary least squares (OLS), maximum likelihood estimation (MLE) under a Normal error distribution, and MLE under a Student's t error distribution.

OLS. The OLS estimates are

$$\hat{\alpha} = 1.560740, \quad \hat{\beta} = 2.990757.$$

The corresponding standard errors are

$$SE(\hat{\alpha}) = 0.096269, \quad SE(\hat{\beta}) = 0.096640.$$

The estimated error standard deviation is

$$\hat{\sigma} = 1.351401.$$

Normal MLE. Under the assumption of Normally distributed errors, the maximum likelihood estimates are

$$\hat{\alpha} = 1.560740, \quad \hat{\beta} = 2.990757,$$

with fitted error standard deviation

$$\hat{\sigma} = 1.344627.$$

The corresponding corrected Akaike Information Criterion is

$$AICc = 692.144523.$$

Student's t MLE. Under the assumption of Student's t distributed errors, the maximum likelihood estimates are

$$\hat{\alpha} = 1.547979, \quad \hat{\beta} = 3.019179.$$

The fitted error parameters are

$$\hat{\sigma} = 0.935509, \quad \hat{\nu} = 3.632161,$$

where ν denotes the degrees of freedom of the Student's t distribution. The corresponding corrected Akaike Information Criterion is

$$AICc = 665.327487.$$

Model Comparison. The Normal and Student's t models can be compared using AICc, where a lower AICc indicates a preferred model. The results are

$$AICc_{\text{Normal}} = 692.144523,$$

and

$$AICc_{\text{Student's } t} = 665.327487.$$

Since the Student's t model has a substantially lower AICc than the Normal model, the Student's t model is preferred. This suggests that the heavier-tailed Student's t error distribution provides a better fit to the data than the Normal error distribution.

Reconcile

- c. Compare the three estimates of β . Did the violation move the slope as much as you predicted?

The three estimates of the slope are

$$\hat{\beta}_{\text{OLS}} = 2.990757, \quad \hat{\beta}_{\text{Normal}} = 2.990757, \quad \hat{\beta}_t = 3.019179.$$

The OLS and Normal MLE estimates are identical, while the Student's t estimate is only slightly larger. The difference between the OLS and Student's t estimates is

$$3.019179 - 2.990757 = 0.028422.$$

When the error distribution changes from a Normal distribution to a Student's t distribution, the estimate of β changes only slightly. Specifically, $\hat{\beta}$ increases from 2.990757 under the Normal model to 3.019179 under the Student's t model. Therefore, the violation of the Normal-error assumption did not move the slope as much as expected. The main effect of the non-normal errors is on the distribution of the residuals rather than on the estimated slope.

- d. The AICc prefers one model by a wide margin, yet the slopes are close together. What, specifically, is the rejected model getting wrong, if not the slope?

The AICc strongly prefers the Student's t model:

$$AICc_{\text{Normal}} = 692.144523, \quad AICc_t = 665.327487.$$

Although the AICc values are substantially different, the estimated slopes are very close. This means that the Normal model does not get the relationship between x and y very wrong. Instead, the main problem is that the Normal model does not describe the error distribution well.

In particular, the Student's t model estimates

$$\hat{\nu} = 3.632161,$$

which indicates that the errors have heavy tails. The Normal distribution has thinner tails, so it gives too little probability to extreme residuals. In contrast, the Student's t distribution can better capture these extreme observations. Therefore, the Normal model is mainly getting the tail behavior of the errors wrong, rather than the slope.

- e. Compare the 95% and the 99.5% quantiles of the fitted Normal error against those of the fitted t error. One is wider at 95% and the other is wider at 99.5%. Explain why, and say which one you would want if you were setting a capital buffer.

We compare the 95% and 99.5% quantiles of the fitted Normal and Student's t errors. The fitted Normal model has a larger scale parameter, $\hat{\sigma}_N = 1.344627$, while the Student's t model has a smaller scale parameter, $\hat{\sigma}_t = 0.935509$. This means that the Student's t distribution is more concentrated around its center.

However, the two distributions have different tail behavior. The Normal distribution has relatively thin tails, so it assigns relatively little probability to very large residuals. In contrast, the Student's t distribution has much heavier tails. In this case, the estimated degrees of freedom is only

$$\hat{\nu} = 3.632161,$$

and a smaller degrees of freedom parameter produces heavier tails. Therefore, the Student's t distribution assigns more probability to extreme residuals.

As a result, at the 95% level, the Student's t distribution can be narrower than the Normal distribution because its center is more concentrated. So at the 95% level, the Normal distribution is wider than the Student's t distribution. However, at the more extreme 99.5% level, the heavy tails of the Student's t distribution become more important. Although the Student's t distribution is more concentrated around the center, its tails extend much farther, making it wider than the Normal distribution at the 99.5% level.

If I were setting a capital buffer, I would prefer the Student's t model, particularly at the 99.5% level. A capital buffer is intended to protect against rare and severe losses, so accurately capturing the extreme tail is more important than fitting the center of the distribution alone. The Student's t model therefore provides a more appropriate and conservative description of the extreme risk in these data.

3 Pearson Against Spearman (1.5 points)

`problem3.csv` holds four series, x_1 through x_4 .

Predict

Plot every pair.

- a. Which pair do you expect the Pearson and Spearman coefficients to disagree on most? Which pairs do you expect them to agree on? Say why in each case.

Here is the figure of the six pairs:

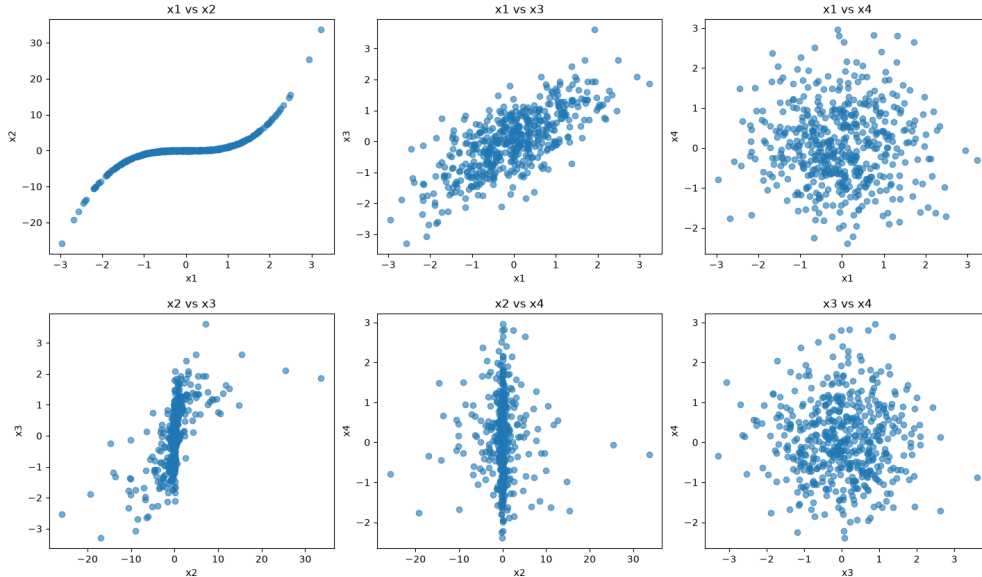


Figure 2: Six pairs of observations.

I expect the Pearson and Spearman coefficients to agree more closely for x_1 vs. x_3 because when x_1 gets larger, x_3 also tends to get larger. The relationship appears to be approximately linear, so Pearson and Spearman should give similar results.

I expect the Pearson and Spearman coefficients to disagree the most for x_1 vs. x_2 because the relationship appears to be nonlinear. As x_1 increases, x_2 generally increases, but not in a linear pattern. Therefore, Spearman may show a stronger relationship than Pearson because Spearman measures monotonic association rather than only linear association.

Fit and reconcile

- b. Compute both correlation matrices. Report the pair with the largest gap.

The pair with the largest gap between the Pearson and Spearman correlations is x_1 vs. x_2 , with a difference of 0.1728. The Pearson and Spearman correlation matrices show that the Spearman correlation (0.9719) is substantially higher than the Pearson correlation (0.7991) for this pair.

- c. Explain the mechanism that produces the gap. Which of the two coefficients is the honest description of that pair, and what question is the other one answering?

The gap occurs because the relationship between x_1 and x_2 is nonlinear but monotonic. As x_1 increases, x_2 generally increases, but the relationship does not follow a straight line. Spearman captures this monotonic relationship by using ranks, while Pearson measures only linear association. Therefore, Spearman is the more honest description if the goal is to describe the overall monotonic relationship between x_1 and x_2 . Pearson is answering a different question: how strong is the linear relationship between x_1 and x_2 ?

4 Conditional Distributions (1.5 points)

`problem4.csv` holds two correlated series, x_1 and x_2 .

Predict

Compute the sample covariance matrix, but nothing else yet.

- a. Write down the conditional variance of x_2 given x_1 using the partitioned result from Week 2, in the blocks of Σ . Say which of your series is which block. If you learn the value of x_1 , by what factor does your uncertainty about x_2 fall? Give that factor as a formula in the same blocks, and then as a number for this data.

Let the covariance matrix be partitioned as from Week 2:

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}.$$

Here,

$$\Sigma_{11} = \text{Var}(X_1), \quad \Sigma_{22} = \text{Var}(X_2),$$

and

$$\Sigma_{12} = \text{Cov}(X_1, X_2), \quad \Sigma_{21} = \text{Cov}(X_2, X_1).$$

For this data, after computing, the covariance matrix is

$$\Sigma = \begin{pmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{pmatrix}.$$

Thus,

$$\Sigma_{11} = 1.099875, \quad \Sigma_{22} = 4.104749,$$

and

$$\Sigma_{12} = \Sigma_{21} = 1.696902.$$

Conditional Variance of X_2 Given X_1

Since X_1 is the regressor, we can write the regression decomposition as

$$X_2 = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(X_1 - \mu_1) + \varepsilon.$$

The coefficient

$$\Sigma_{21}\Sigma_{11}^{-1}$$

is the regression coefficient for regressing X_2 on X_1 . For scalar variables,

$$\beta = \frac{\text{Cov}(X_2, X_1)}{\text{Var}(X_1)} = \Sigma_{21}\Sigma_{11}^{-1}.$$

If X_1 is observed, the conditional mean

$$\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(X_1 - \mu_1)$$

is determined. Therefore, the remaining uncertainty about X_2 comes from the residual ε . Hence,

$$\text{Var}(X_2 | X_1) = \text{Var}(\varepsilon).$$

Using the partitioned covariance formula from Week 2, we could get the conditional variance of X_2 given X_1 is

$$\boxed{\text{Var}(X_2 | X_1) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}.$$

Reduction in Uncertainty

Before observing X_1 , the variance of X_2 is

$$\text{Var}(X_2) = \Sigma_{22}.$$

After observing X_1 , the remaining variance is

$$\text{Var}(X_2 | X_1) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}.$$

Therefore, the fraction of the original variance that remains is

$$\boxed{\text{Fraction remaining} = \frac{\Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}{\Sigma_{22}}}.$$

The fraction of the original variance that is reduced is

$$1 - \frac{\Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}{\Sigma_{22}}.$$

Simplifying,

$$\boxed{\text{Fraction reduced} = \frac{\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}{\Sigma_{22}}}.$$

For scalar variables, since

$$\Sigma_{12} = \Sigma_{21}$$

and

$$\Sigma_{11}^{-1} = \frac{1}{\Sigma_{11}},$$

the fraction reduced can also be written as

$$\frac{\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}{\Sigma_{22}} = \frac{\Sigma_{12}^2}{\Sigma_{11}\Sigma_{22}} = \rho^2.$$

Numerical Calculation

For this data,

$$\beta = \frac{\Sigma_{21}}{\Sigma_{11}} = \frac{1.696902}{1.099875} \approx 1.5428.$$

The conditional variance is

$$\begin{aligned}\text{Var}(X_2 | X_1) &= \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12} \\ &= 4.104749 - \frac{(1.696902)^2}{1.099875} \\ &\approx 1.486746.\end{aligned}$$

Thus,

$$\boxed{\text{Var}(X_2 | X_1) \approx 1.486746}.$$

The fraction of the original variance that remains is

$$\begin{aligned}\text{Fraction remaining} &= \frac{\Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}{\Sigma_{22}} \\ &= \frac{1.486746}{4.104749} \\ &\approx 0.3622.\end{aligned}$$

Therefore,

$$\boxed{\text{Fraction of variance remaining} \approx 36.22\%}.$$

The fraction of the original variance that is reduced is

$$\begin{aligned}\text{Fraction reduced} &= \frac{\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}{\Sigma_{22}} \\ &= 1 - 0.3622 \\ &\approx 0.6378.\end{aligned}$$

Therefore,

$$\boxed{\text{Fraction of variance reduction} \approx 63.78\%}.$$

Hence, if we learn the value of X_1 , the variance-based uncertainty about X_2 falls to approximately 36.22% of its original level, corresponding to a reduction of approximately 63.78%.

- b. Under the multivariate normal, does that factor depend on the value of x_1 you observed? Point at the term in the formula that settles it, and answer before you look at the data.

Under the multivariate normal distribution, based on part (a), the conditional variance of X_2 given X_1 is

$$\text{Var}(X_2 | X_1) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}.$$

The key point is that this expression contains only the covariance matrix Σ and does not contain the observed value of x_1 . Therefore, the factor by which uncertainty falls does not depend on the particular value of x_1 that is observed.

Fit and Reconcile.

- c. Write down the conditional mean of x_2 given x_1 , in the same blocks of Σ . Identify the coefficient on x_1 and say what it is in regression terms. Then plot the conditional expectation across the range of x_1 , with a 95% band around it, over a scatter plot of the data.

First, partition the covariance matrix as

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}.$$

Since we only have two series, the blocks are

$$\Sigma_{11} = \text{Var}(X_1),$$

$$\Sigma_{22} = \text{Var}(X_2),$$

$$\Sigma_{12} = \text{Cov}(X_1, X_2),$$

and

$$\Sigma_{21} = \text{Cov}(X_2, X_1).$$

For jointly normally distributed X_1 and X_2 , the conditional mean of X_2 given $X_1 = x_1$ is

$$\mathbb{E}[X_2 \mid X_1 = x_1] = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - \mu_1).$$

Because X_1 and X_2 are both scalars, we can write the conditional mean in the more intuitive form:

$$\mathbb{E}[X_2 \mid X_1 = x_1] = \mu_2 + \frac{\Sigma_{21}}{\Sigma_{11}}(x_1 - \mu_1)$$

Equivalently,

$$\mathbb{E}[X_2 \mid X_1 = x_1] = \alpha + \beta x_1,$$

where

$$\beta = \frac{\Sigma_{21}}{\Sigma_{11}}$$

and

$$\alpha = \mu_2 - \beta\mu_1.$$

The coefficient on x_1 is

$$\beta = \frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)}$$

which is exactly the OLS regression slope coefficient when regressing X_2 on X_1 :

$$X_2 = \alpha + \beta X_1 + \varepsilon.$$

Therefore, the coefficient on x_1 is $\Sigma_{21}\Sigma_{11}^{-1}$. In the scalar case, this is

$$\frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)},$$

which is exactly the OLS regression slope from regressing X_2 on X_1 .

The plot of the conditional expectation across the range of x_1 , with a 95% band around it, over a scatter plot of the data:

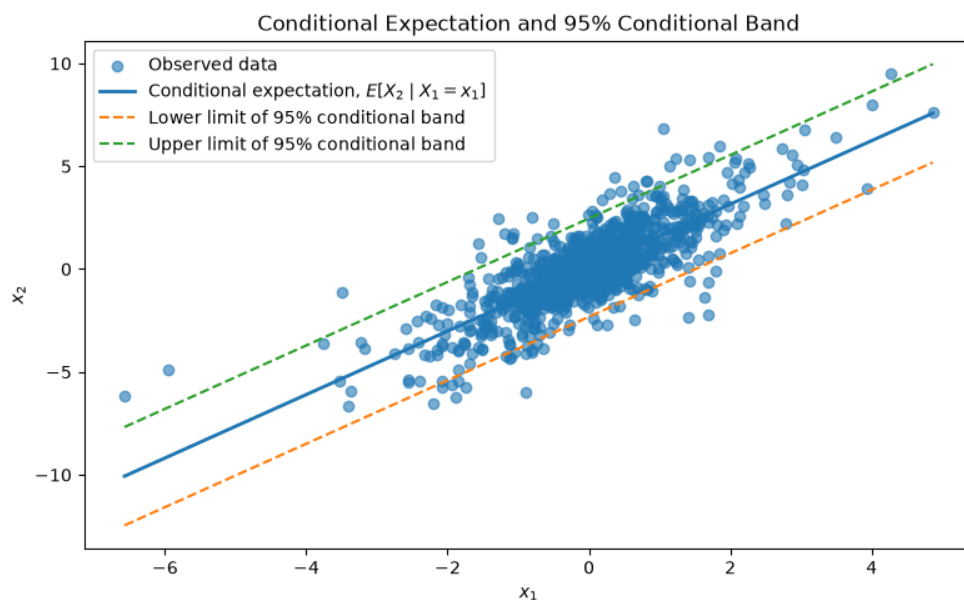


Figure 3: Conditional expectation and 95% conditional prediction band.

d. Compute the fraction of observations that fall inside the band. Report it.

93.5% of the observations fall inside the 95% band.

e. Now split the sample by how far x_1 sits from its mean—inside one standard deviation, between one and two standard deviations, and beyond two standard deviations—and compute the coverage within each bucket. Report all three.

- Coverage within one standard deviation: 95.65%
- Coverage between one and two standard deviations: 85.79%

- Coverage beyond two standard deviations: 90.20%

f. The coverage is not flat across those buckets. Explain what that tells you about the data, and identify which assumption behind the constant-width band has failed. Note carefully which parts of your answer to part (c) survive the failure and which do not.

The coverage is not approximately constant across the three buckets: it is 95.65% within one standard deviation, 85.79% between one and two standard deviations, and 90.20% beyond two standard deviations. This suggests that the conditional distribution of X_2 changes with the value of X_1 , particularly away from its mean. In other words, the conditional variance of X_2 is not constant across values of X_1 . Therefore, the assumption of homoskedasticity (constant conditional variance) underlying the constant-width band has failed. The conditional mean specification may still be valid, but the constant conditional variance assumption is not. Therefore, the conditional mean survives, while the constant-width prediction band does not.

5 Identifying an AR or MA Order (1.5 points)

problem5.csv holds a single series, x .

Predict

Plot the series, its ACF, and its PACF.

- a. Which of the two functions cuts off, and which decays? What order does that imply, and is it an AR or an MA process? Commit to a specific answer before fitting.

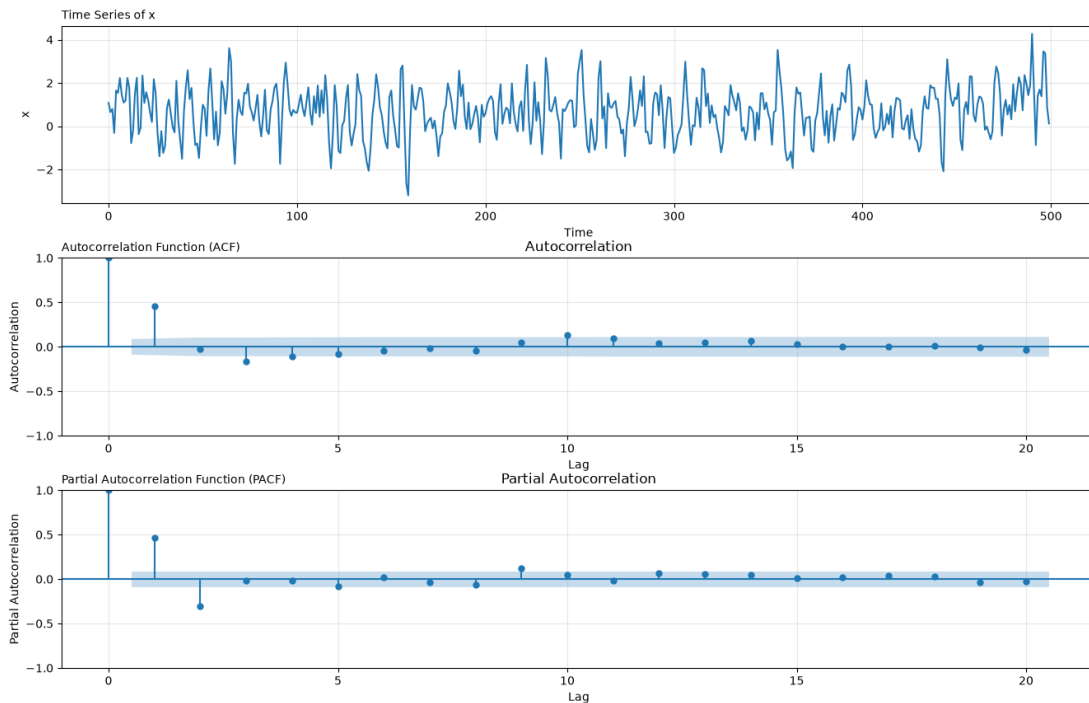


Figure 4: Time series of x , its autocorrelation function (ACF), and partial autocorrelation function (PACF).

The ACF cuts off after lag 1, while the PACF gradually decays toward zero. Based on the standard ACF/PACF identification rule, this pattern indicates a moving average process of order 1. Therefore, before fitting the model, I predict that the series follows an $(MA(1))$ process.

- b. State the rule you used. Cite the significance band you judged “cuts off” against.

The rule I used is that an $MA(q)$ process has an ACF that cuts off after lag q , while its PACF decays gradually. In contrast, an $AR(p)$ process has a PACF that cuts off after lag p , while its ACF decays.

I judged a function to have “cut off” when the autocorrelation or partial autocorrelation became statistically insignificant and remained within the approximate 95% significance band,

$$\pm \frac{1.96}{\sqrt{n}}.$$

For this series, the ACF is significant through lag 1 and then falls within the significance band, while the PACF gradually decays. This leads to the prediction of an $MA(1)$ process.

Fit and reconcile.

- c. Fit $AR(1)$, $AR(2)$, $AR(3)$, $MA(1)$, $MA(2)$, and $MA(3)$. Report the AICc for each.

Model	AICc
$AR(1)$	1416.389682
$AR(2)$	1368.371829
$AR(3)$	1368.467605
$MA(1)$	1389.084281
$MA(2)$	1381.231619
$MA(3)$	1374.116525

- d. Which model does the AICc select? Did it match your prediction?

The AICc selects the $AR(2)$ model because it has the smallest AICc value, 1368.371829, among the six candidate models. This result does not match my prediction based on the ACF and PACF analysis before. I predicted a $MA(1)$ process before because the ACF appeared to cut off after lag 1 while the PACF gradually decayed. The ACF/PACF rule is only a heuristic for identifying a possible model. Because the ACF and PACF are estimated from a finite sample, random sampling variation can cause the observed patterns to differ from their theoretical patterns. Therefore, the ACF/PACF suggested an $MA(1)$ model, but the AICc comparison favored $AR(2)$.

- e. Compare the $AR(2)$ and $AR(3)$ fits. The third coefficient is small. Explain what AICc is trading off when it rejects the larger model, and why R^2 would not have made the same choice.

The $AR(3)$ model has a slightly larger AICc than the $AR(2)$ model:

$$AICc_{AR(2)} = 1368.371829, \quad AICc_{AR(3)} = 1368.467605.$$

The third AR coefficient is small, so adding the third lag provides only a small improvement in model fit. Importantly, this does not mean that the $AR(3)$ model is necessarily a bad model; rather, its

improvement over AR(2) is too small to justify the additional model complexity. AICc trades off goodness of fit against model complexity by penalizing the addition of extra parameters. In this case, the improvement from adding the third coefficient is not large enough to offset the complexity penalty, so AICc favors the simpler AR(2) model.

R^2 would not make the same choice because R^2 does not penalize the addition of extra predictors. Adding another lag can only increase or leave R^2 unchanged, even when the additional coefficient contributes very little to the model. Therefore, R^2 would tend to favor AR(3) simply because it has an additional predictor and slightly better in-sample fit. In contrast, AICc asks whether the improvement in fit is large enough to justify the additional complexity. Here, it is not, so AICc rejects the larger AR(3) model in favor of the simpler AR(2) model.